

H524 Introduction to Biostatistics

Week 5 Assignment - ANSWER KEY

Fall 2025

Total Points: 65 (+ 5 bonus)

Instructor Use Only

Answer Key - Quick Reference

Part 1: Hypothesis Testing Fundamentals (18 points)

Q#	Answer	Pts	Topic
1	D	3	Null hypothesis definition
2	C	3	Alternative hypothesis (one-sided, lower)
3	C	4	P-value interpretation
4	B	2	Decision rule
5	A	3	T-statistic = 1.25
6	A	3	P-value = 0.2234

Part 2: Type I and Type II Errors (14 points)

Q#	Answer	Pts	Topic
7	C	3	Type I error (false positive)
8	B	3	Type II error (false negative)
9	D	3	Error trade-off
10	B	2	Statistical power
11a	5.18 or 5.181	5	Blood pressure t-statistic
11b	0.0001 or 3.17e-05	5	Blood pressure p-value

Part 3: Hypothesis Testing with R (10 points)

Q#	Answer	Pts	Topic
12a	-2.71 or -2.7105	5	Cholesterol t-statistic
12b	0.0050 or 0.004967	5	Cholesterol p-value

Part 4: Two-Sample Tests (14 points)

Q#	Answer	Pts	Topic
13	B	3	Two-sample application
14	A	4	Two-sample calculation
15	D	2	R code for two-sample test

Detailed Solutions

Part 1: Hypothesis Testing Fundamentals

Question 1 (3 points): Null Hypothesis Definition

Answer: D

H_0 represents the status quo or no effect; it is the hypothesis we assume true until we have sufficient evidence to reject it. This is the fundamental principle of hypothesis testing - we start by assuming no effect and require strong evidence to reject this assumption.

Question 2 (3 points): Alternative Hypothesis Forms

Answer: C

Since the researcher wants to test if the drug LOWERS blood pressure compared to 140 mmHg, the appropriate alternative hypothesis is $H_a: \mu < 140$ (one-sided, lower tail). This is a directional hypothesis testing for a decrease.

Question 3 (4 points): P-value Interpretation

Answer: C

The p-value is the probability of observing data as extreme as (or more extreme than) what we observed, assuming the null hypothesis is true. It is NOT the probability that H_0 is true given our data.

Question 4 (2 points): Decision Rule

Answer: B

Since p-value (0.032) $< \alpha$ (0.05), we reject H_0 . We have sufficient evidence against the null hypothesis at the 5% significance level.

Question 5 (3 points): T-statistic Calculation

Answer: A (t = 1.25)

Solution:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{52 - 50}{8/\sqrt{25}} = \frac{2}{8/5} = \frac{2}{1.6} = 1.25$$

Question 6 (3 points): P-value from t-statistic

Answer: A (p-value = 0.2234)

Solution: For $t = 1.25$ with $df = 24$ (two-sided test):

```
2 * pt(1.25, df = 24, lower.tail = FALSE)
# Result: 0.2234
```

Since this is a two-sided test, we need to multiply the one-tail probability by 2.

Part 2: Type I and Type II Errors

Question 7 (3 points): Type I Error

Answer: C

Type I error occurs when we reject the null hypothesis when it is actually true (false positive). The probability of Type I error is controlled by the significance level α .

Question 8 (3 points): Type II Error

Answer: B

This is a Type II error (β) - failing to reject a false null hypothesis. The drug is actually effective (null is false), but the test failed to detect it (failed to reject the null).

Question 9 (3 points): Error Trade-off

Answer: D

When you decrease α (become more conservative about rejecting H_0), you increase β (the probability of failing to reject when you should). Making it harder to reject H_0 increases the chance of missing a true effect.

Question 10 (2 points): Statistical Power

Answer: B

Power = $1 - \beta$; it's the probability of correctly rejecting a false null hypothesis (detecting a true effect when it exists). Higher power means better ability to detect true effects.

Question 11 (10 points): Blood Pressure Hypothesis Test

Data:

```
bp_data <- c(118, 124, 130, 128, 122, 135, 119, 126, 132, 120,
            125, 129, 121, 127, 123, 131, 120, 128, 124, 126)
```

R Code:

```
result <- t.test(bp_data, mu = 120, alternative = "greater")
result$statistic # t-statistic
result$p.value   # p-value
```

Answers:

a) **t-statistic:** 5.18 (or 5.181 for more precision)

b) **p-value:** 0.0001 (or 3.17e-05 for exact value)

Interpretation: With such a small p-value (much less than 0.05), we have very strong evidence that the mean blood pressure is significantly higher than 120 mmHg.

Part 3: Hypothesis Testing with R

Question 12 (10 points): Cholesterol Hypothesis Test

Given: $n = 40$, $\bar{x} = 185$ mg/dL, $s = 35$ mg/dL

Hypotheses: $H_0: \mu = 200$ vs $H_a: \mu < 200$ (one-sided, lower tail)

Manual Calculation:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{185 - 200}{35/\sqrt{40}} = \frac{-15}{35/6.325} = \frac{-15}{5.534} = -2.71$$

R Code:

```
# Manual calculation
n <- 40
xbar <- 185
s <- 35
mu0 <- 200

t_stat <- (xbar - mu0) / (s / sqrt(n))
p_value <- pt(t_stat, df = n - 1)

# Or using t.test
# Note: We don't have the actual data, so we use manual calculation
# If we had data: t.test(chol_data, mu = 200, alternative = "less")
```

Answers:

- a) **t-statistic:** -2.71 (or -2.7105 for more precision)
- b) **p-value:** 0.0050 (or 0.004967 for more precision)

Interpretation: At $\alpha = 0.01$, the p-value (0.0050) is less than 0.01, so we reject H_0 . We have strong evidence that the drug reduces cholesterol to below 200 mg/dL on average. The company's claim is supported by the data.

Part 4: Two-Sample Tests

Question 13 (3 points): Two-Sample Test Application

Answer: B

Comparing mean recovery times between patients receiving Drug A versus Drug B requires a two-sample t-test because we're comparing two independent groups. The other options all compare a sample mean to a known value (requiring one-sample tests).

Question 14 (4 points): Two-Sample T-test

Answer: A

Given:

- Treatment A: $n_1 = 30$, $\bar{x}_1 = 145$, $s_1 = 15$
- Treatment B: $n_2 = 35$, $\bar{x}_2 = 138$, $s_2 = 18$

R Code:

```
# Create data from summary statistics (approximation)
group_a <- rnorm(30, mean = 145, sd = 15)
group_b <- rnorm(35, mean = 138, sd = 18)

# Two-sample t-test
result <- t.test(group_a, group_b, var.equal = FALSE)
result$statistic
result$p.value
```

Result: $t \approx 1.71$, p-value ≈ 0.0922

Decision: Since p-value (0.0922) $> \alpha$ (0.05), we fail to reject H_0 . There is insufficient evidence to conclude that the two treatments have different mean values at the 5% significance level.

Question 15 (2 points): R Code for Two-Sample Test

Answer: D

The correct syntax is `t.test(group_a, group_b)`. This performs a two-sample t-test comparing the means of the two groups.

Bonus Question - Grading Guidelines

Question 16: AI Verification and Learning (5 bonus points)

Grading Notes:

Part a: (1 point) Student should provide actual AI prompt and response

- Clear, specific prompt: 0.5 points
- AI response documented: 0.5 points

Part b: (2 points) Student should verify AI answer step-by-step

- Step-by-step verification with calculations: 1 point
- Correctly identifying accuracy/errors: 1 point

Part c: (2 points) Student should evaluate AI's explanation of one-sided vs. two-sided tests

- Providing AI's explanation: 0.5 points
- Thoughtful evaluation of quality and clarity: 1 point
- Discussion of strengths/weaknesses: 0.5 points

Look for:

- Clear documentation of AI interaction
- Critical thinking about AI responses
- Understanding of hypothesis testing concepts
- Ability to verify and evaluate AI output
- Recognition that AI can make mistakes

Grading Summary

Section	Points	Score
Part 1: Fundamentals (Q1-Q6)	18	
Part 2: Errors and Power (Q7-Q11)	14	
Part 3: R Implementation (Q12)	10	
Part 4: Two-Sample Tests (Q13-Q15)	14	
Subtotal	65	
Bonus: AI Learning (Q16)	+5	
Total Possible	70	

Common Student Errors

- **Q5:** Forgetting to divide by $\sqrt{n}$ in the denominator
- **Q6:** Using one-tail p-value instead of two-tail (forgetting to multiply by 2)
- **Q11:** Using wrong alternative (“two.sided” instead of “greater”)
- **Q12:** Using wrong alternative (“greater” instead of “less”)
- **Q14:** Attempting to reject at $\alpha = 0.05$ when p-value is close (0.0922)

Note to Graders

Computational Verification:

- All numerical answers in this key have been verified using R (see `verify_answers.R` in this directory)
- For Questions 11 and 12: Accept answers within reasonable rounding error
 - Q11a: Accept 5.18, 5.181, or 5.2
 - Q11b: Accept 0.0001, 3.17e-05, or <0.001
 - Q12a: Accept -2.71, -2.7105, or -2.7
 - Q12b: Accept 0.005, 0.00497, or 0.0050

Partial Credit Guidelines:

- **Q11 & Q12 (Computational):** Award partial credit for correct approach even if calculation is wrong
 - Correct code structure with minor errors: 70% credit
 - Correct formula but computational mistake: 80% credit
 - Correct method but wrong R function arguments: 60% credit
- **Bonus Question (Q16):** Use detailed rubric on pages 5-6; look for critical thinking, not just AI copy-paste

Academic Integrity:

- Watch for identical code/answers across students (especially Q11-12)
- Bonus question should show student’s OWN evaluation, not just AI output
- Flag suspicious patterns for instructor review

Common Mistakes to Watch For:

- Using `alternative = "two.sided"` when problem specifies one-sided test
- Confusing one-tail and two-tail p-values (Q6)
- Wrong direction for alternative hypothesis (Q11 should be “greater”, Q12 should be “less”)